

ANGOULEME-COGNAC AP, France

WMO: 074200

Lat: 45.7279N

Lon: 0.2205E

Elev: 436

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	26.9	14.4	11.6	26.6	20.9	15.9	30.2	23.3	50.2	20.9	48.8	7.6	10	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.3	91.4	69.0	87.4	67.5	83.9	66.3	71.0	85.7	69.4	83.6	67.9	80.8	7.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	98.1	76.4	64.6	92.6	74.4	63.0	87.6	72.5	35.0	85.8	33.7	83.6	32.5	80.6	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	17.7	15.7	DB	20.4	98.0	5.9	2.7	16.2	99.9	12.8	101.5	9.5	103.0	5.3	104.9	(4)
(5)			WB	19.3	74.3	5.7	2.4	15.2	76.0	11.8	77.3	8.6	78.7	4.4	80.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.1	41.7	42.4	48.8	53.1	58.0	65.2	67.9	68.5	63.6	57.1	49.1	44.4		
	DBStd	11.54	6.51	7.90	6.12	5.97	6.46	7.62	6.11	6.48	6.09	6.87	7.04	6.77		
	HDD50	937	263	224	96	29	9	3	0	0	1	17	102	193		
	HDD65	4080	722	632	502	360	232	88	38	36	95	258	477	640		
	CDD50	2785	6	11	59	123	258	458	555	574	411	236	75	19		
	CDD65	454	0	0	0	3	16	93	128	146	55	13	0	0		
	CDH74	4953	0	0	1	57	206	1064	1450	1566	537	72	0	0		
	CDH80	1713	0	0	0	6	35	383	516	631	133	9	0	0		
(14) Wind	WSAvg	7.5	7.7	8.4	8.8	7.7	7.5	7.2	6.9	6.3	6.7	6.9	7.6	7.8		
(15) Precipitation	PrecAvg	32.4	3.4	2.6	2.5	2.5	2.9	2.4	2.0	2.4	2.4	2.9	3.5	3.1		
	PrecMax	38.3	7.6	5.0	6.7	7.4	5.0	3.8	3.2	5.0	6.0	4.8	6.8	6.7		
	PrecMin	24.7	1.0	0.6	0.8	0.5	0.6	0.3	0.4	0.3	1.1	1.3	1.1	0.9		
	PrecStd	4.2	1.7	1.4	1.5	1.7	1.2	0.9	0.9	1.5	1.2	1.1	1.6	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	64.8	73.2	80.6	84.1	96.2	96.5	95.4	88.1	80.5	68.3	60.4		
		MCWB	52.3	51.3	54.9	60.4	65.9	71.8	70.1	69.3	64.1	62.9	59.7	53.4		
	2%	DB	55.1	59.5	68.2	74.8	79.1	88.1	89.7	91.5	84.1	75.3	64.5	57.4		
		MCWB	51.9	50.4	53.4	57.4	63.0	68.5	69.4	68.3	64.2	62.8	56.8	52.4		
	5%	DB	53.2	55.5	64.4	69.9	75.4	84.3	85.6	87.4	80.2	71.3	61.8	55.4		
		MCWB	50.7	48.7	52.6	55.3	61.8	67.2	67.0	68.1	63.8	62.1	56.3	52.1		
	10%	DB	51.3	52.2	60.4	65.8	71.7	80.5	82.0	82.5	76.8	67.7	59.0	53.7		
		MCWB	48.8	48.0	50.5	53.8	59.6	66.1	65.7	66.2	62.7	60.0	55.6	50.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.6	53.8	56.3	61.0	68.8	73.9	72.8	73.0	69.0	66.9	61.4	55.6		
		MCDB	55.6	61.6	68.0	75.8	80.7	89.1	89.4	88.0	80.0	75.1	66.0	57.2		
	2%	WB	52.8	51.8	54.8	58.7	65.7	70.6	70.6	70.6	67.0	64.6	58.8	54.2		
		MCDB	54.3	56.4	65.2	71.9	76.0	84.2	86.0	85.8	77.8	71.0	61.9	56.5		
	5%	WB	51.0	50.2	53.3	56.4	63.1	68.8	68.7	69.0	65.3	63.0	57.4	52.8		
		MCDB	52.9	53.8	62.3	67.6	72.4	82.4	82.9	83.5	75.3	68.8	60.5	55.3		
	10%	WB	48.8	48.8	51.6	54.5	60.7	66.5	67.0	67.3	63.7	61.2	55.6	51.2		
		MCDB	50.5	52.3	58.1	64.1	68.7	78.7	79.5	79.8	73.0	67.1	59.3	53.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	14.0	17.3	20.1	20.9	22.4	22.7	23.3	21.1	17.5	12.5	11.7		
		MCDBR	11.7	19.3	24.3	25.6	26.2	26.4	29.7	30.1	26.4	21.3	13.9	12.7		
		MCWBR	9.2	12.0	12.9	13.1	13.3	11.6	12.1	11.8	11.1	11.1	8.3	8.8		
	5% WB	MCDBR	10.4	16.6	21.1	23.1	24.3	24.8	27.4	25.5	21.5	17.7	12.6	10.5		
		MCWBR	9.2	11.5	12.2	12.5	14.0	12.2	13.6	12.0	10.9	10.3	8.7	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.319	0.335	0.366	0.393	0.396	0.409	0.399	0.393	0.364	0.361	0.335	0.316		
	taud		2.469	2.412	2.330	2.251	2.295	2.293	2.319	2.351	2.412	2.434	2.453	2.465		
	Ebn at Noon		249	267	272	274	277	272	274	271	271	253	239	237		
	Edh at Noon		22	28	35	41	40	41	39	36	31	27	22	20		
(44) All-Sky Solar Radiation	RadAvg		390	680	1050	1442	1686	1929	1881	1664	1321	821	464	351		
	RadStd		45	83	128	171	146	164	121	128	87	69	57	51		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page